

rooms. These scaffold a student's interactions with peers and teachers in real time.

IIT Bombay has introduced a low-cost laptop, FOSSEE, with more than 150 open source software packages on this system. The Indian government is also spending a lot of money to create good-quality educational content through initiatives such as National Programme on Technology Enhanced Learning (NPTEL) and Virtual Labs. Another important initiative is Spoken Tutorial for IT training.

Simulation software. Simulation tools are used when conducting experiments on a real system would be impossible or impractical; for example, when prototyping and testing a system involves a high cost, a delicate product doesn't support extensive tests, or duration of the experiment in real time is impractical. Simulation tools help to disseminate conceptual and complex ideas to a wide variety of learners using appropriate media and methods to promote understanding. Going beyond demos, simulation provides animated, interactive and game-like environments to enhance analytic skill. Computer simulation increases perception of students.



zSpace 200 provides an immersive, interactive hardware and software platform for students, educators, researchers and corporate trainers (Image courtesy: <https://upload.wikimedia.org>)

Virtual reality and augmented reality. Augmented reality and virtual reality are both related technologies, and using 3D environment or head-mounted display these provide an enhanced or enriched experience of an imaginary reality.

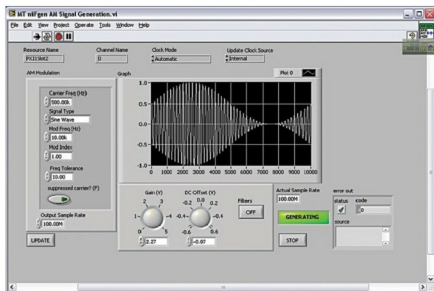
Virtual reality is commonly achieved through a head-mounted display, or handheld controller. It is an artificial simulation of a real-life environment or situation where people can have hands-on practice, like flight simulation for aircraft

pilots. It immerses the users by making them feel like they are experiencing the situation. On the other hand, augmented reality is used more in mobile devices like smart phones, laptops and tablets. Augmented reality-based piano learning apps/trainers are also available in the market. Virtual Reality Modeling Language (VRML) is the coding language used for virtual reality.

Skills can be easily enhanced through virtual and augmented reality-based training. For example, new train drivers can be trained in driving skills using simulators that provide a realistic simulation of any event and allow exercise of control in any crisis situation. Such a training provides the opportunity to safely train the driver in avoiding real-life accidents.

The future of education resides in virtual reality and augmented reality environments where students from around the world converge in a magical hands-on learning experience. Virtual reality creates an immersive audio, visual and spatial learning environment, which helps students to explore and learn at their own pace.

Flipped classrooms. Flipping the classroom is a new approach to education. It reverses the traditional



Amplitude-modulated signal generation using NI-GEN waveform generator—a virtual instrument (Image courtesy: www.ni.com)